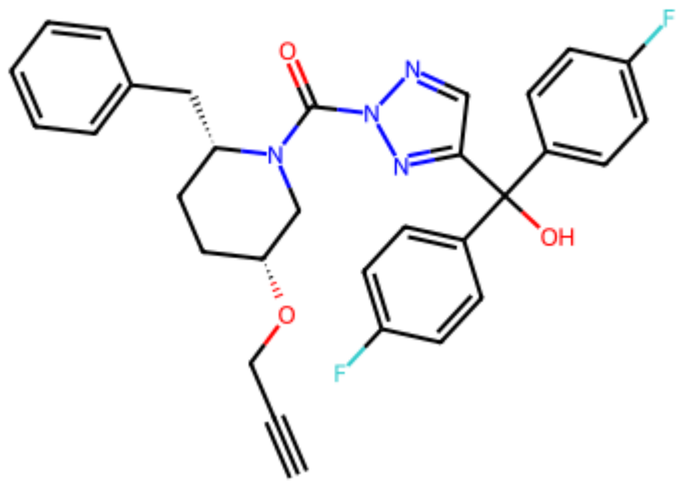
🧪 MCS-Grouped Retrosynthesis Report

Predictions grouped by Maximum Common Substructure (MCS) Tanimoto similarity of the largest reactant fragment — distance threshold = 0.50. All structures are aligned to the product via the MCS core.

Product 1

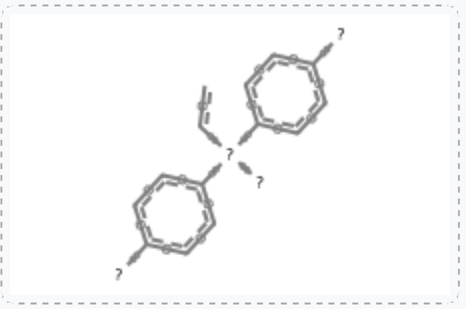
`C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(O)(c3ccc(F)cc3)c3ccc(F)cc3)n2)C1`

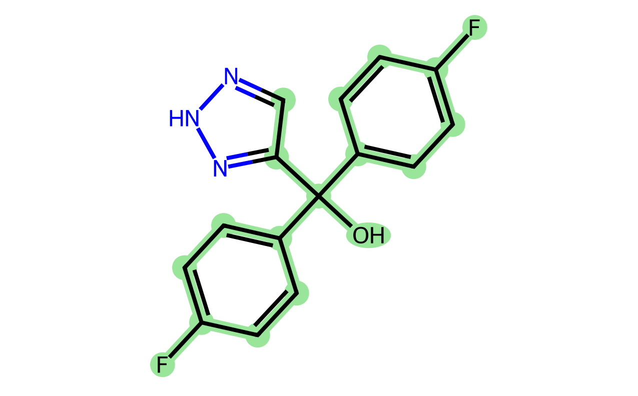
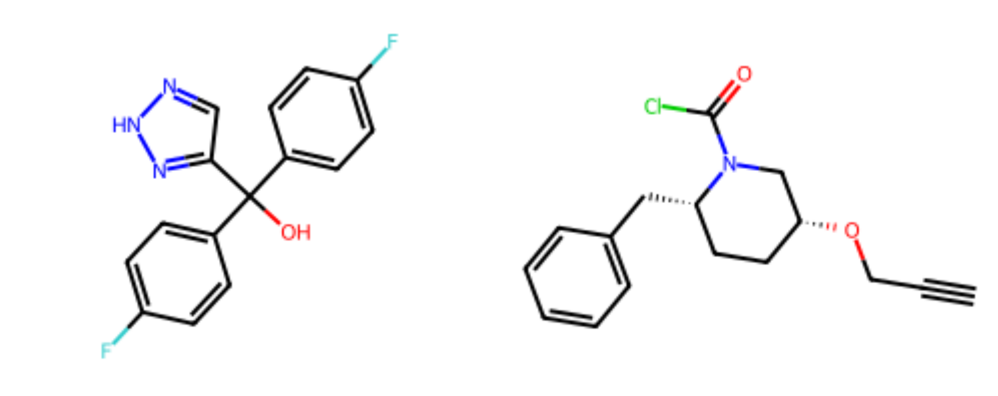
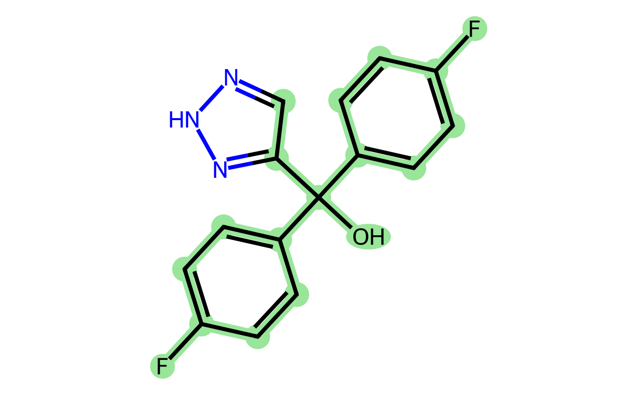
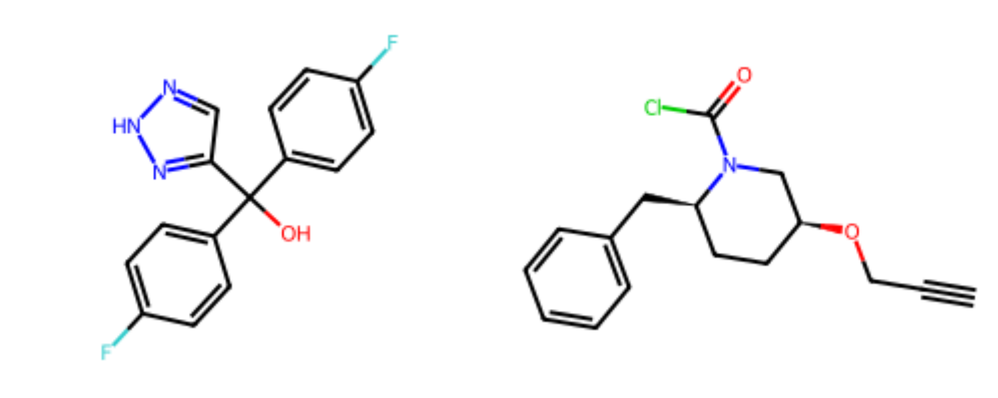
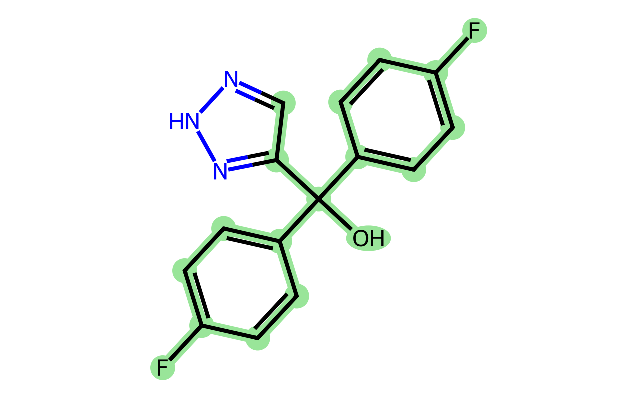
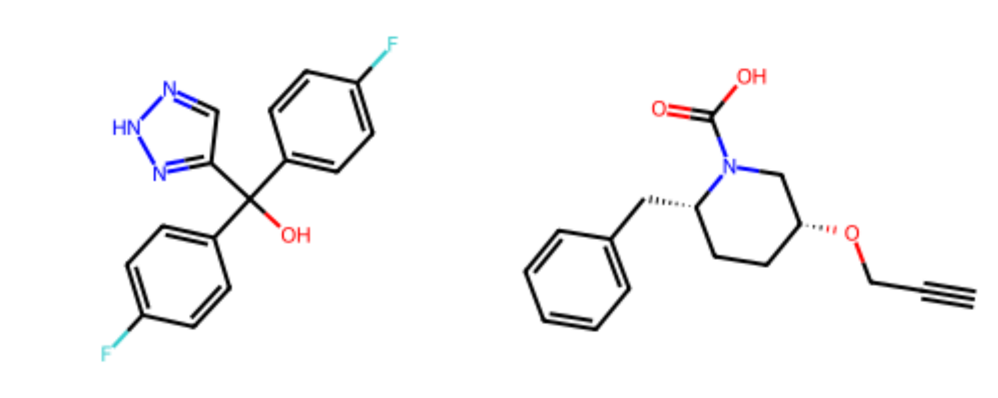


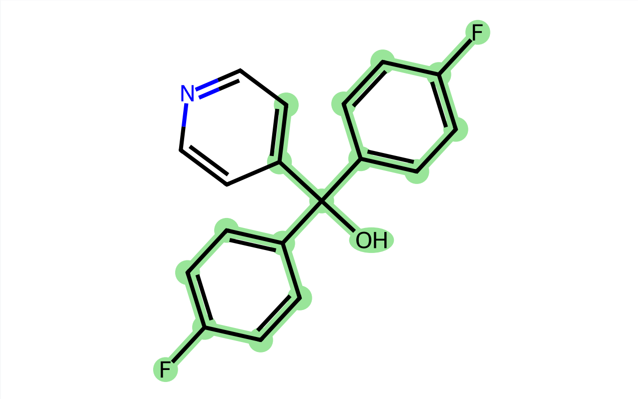
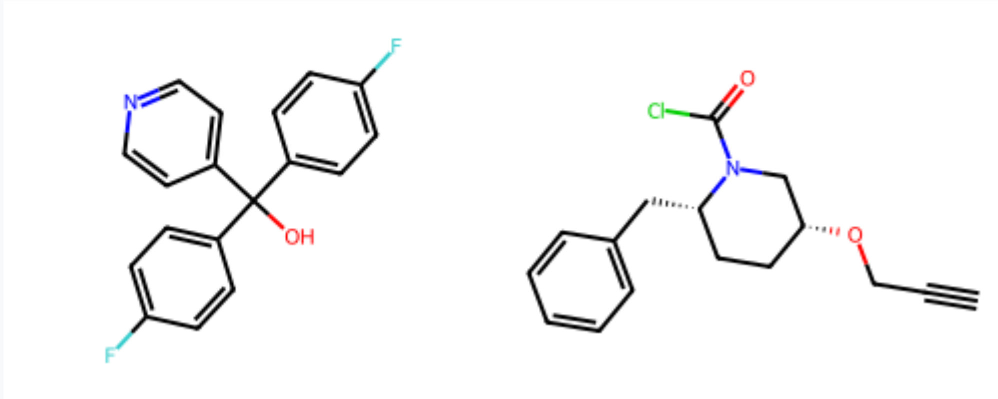
10 predictions → **2** MCS match groups

MCS Match Group 1 (4 predictions) — best score -0.0027

MCS core (18 atoms):

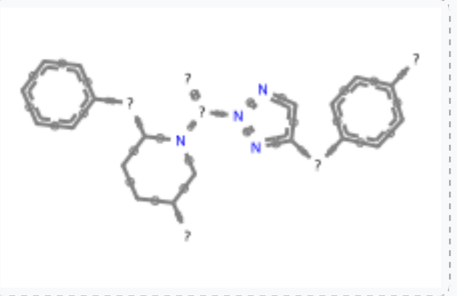


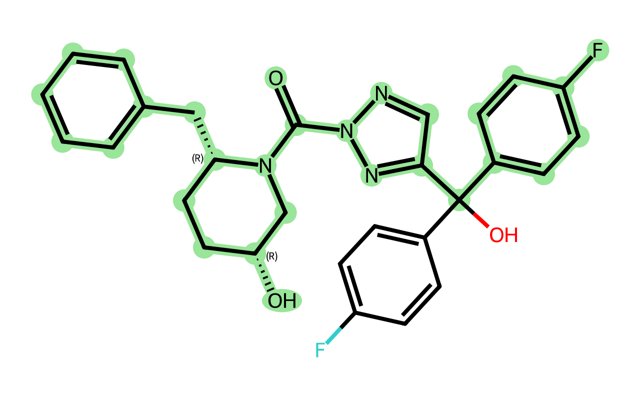
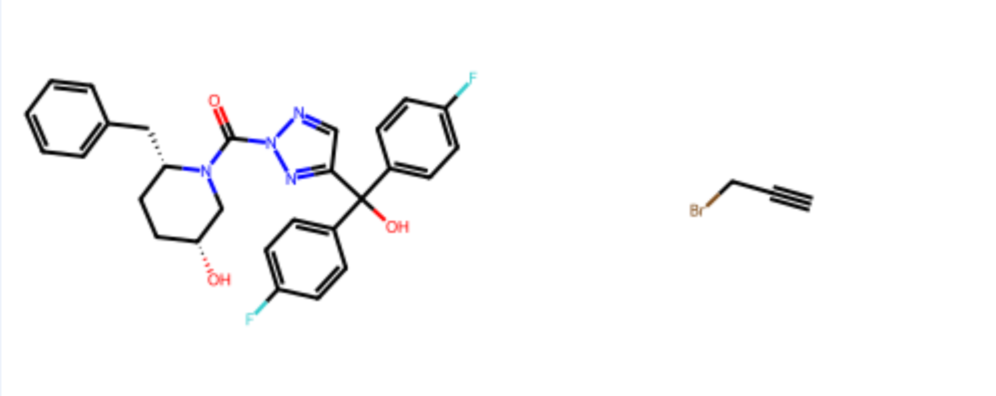
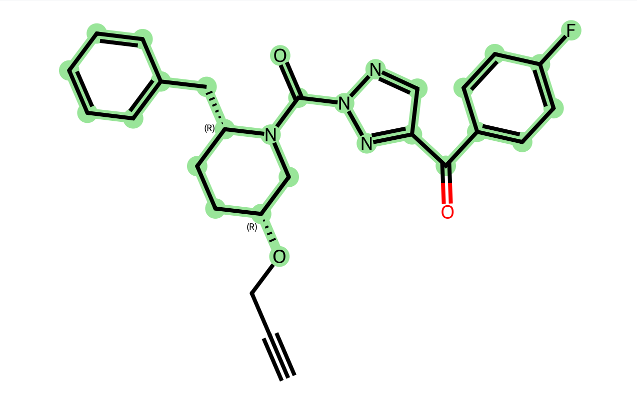
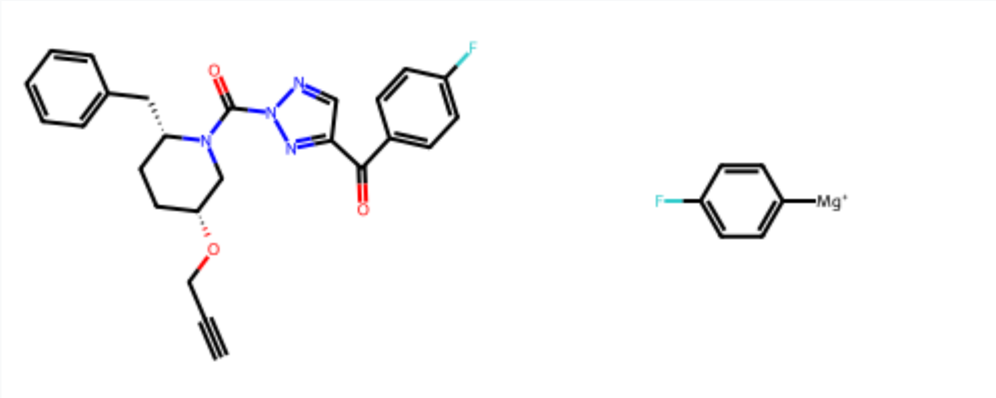
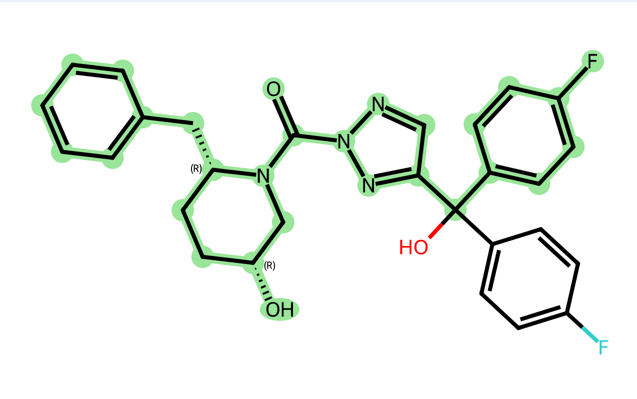
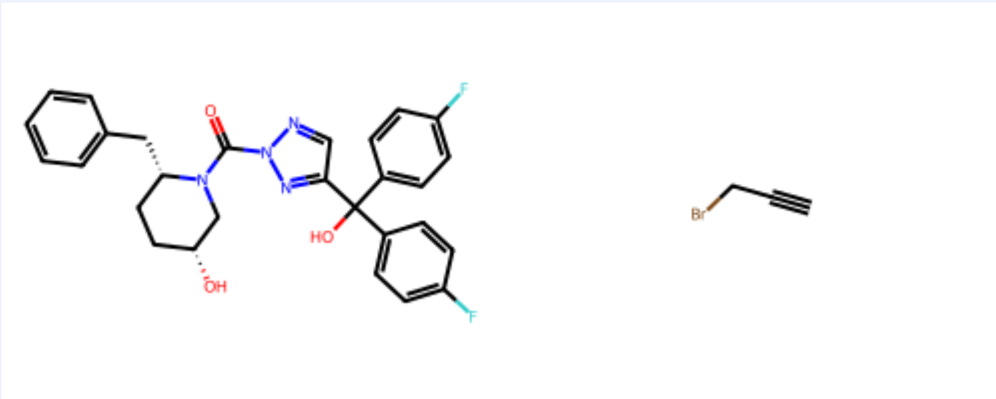
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0027	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)Cl)C1.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1		
Top-3	-0.0455	C#CCO[C@H]1CC[C@@H](Cc2ccccc2)N(C(=O)Cl)C1.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1		
Top-7	-0.0696	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)O)C1.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cn[nH]n1		

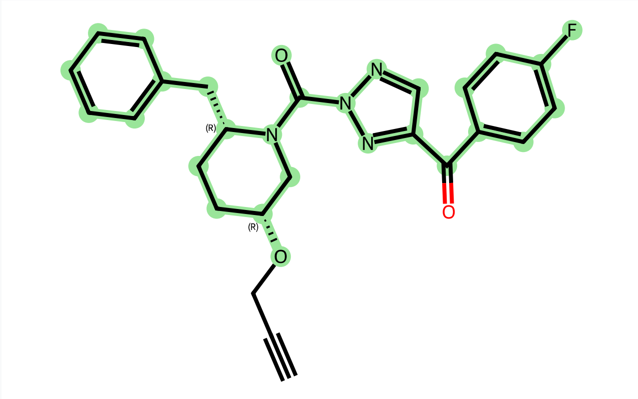
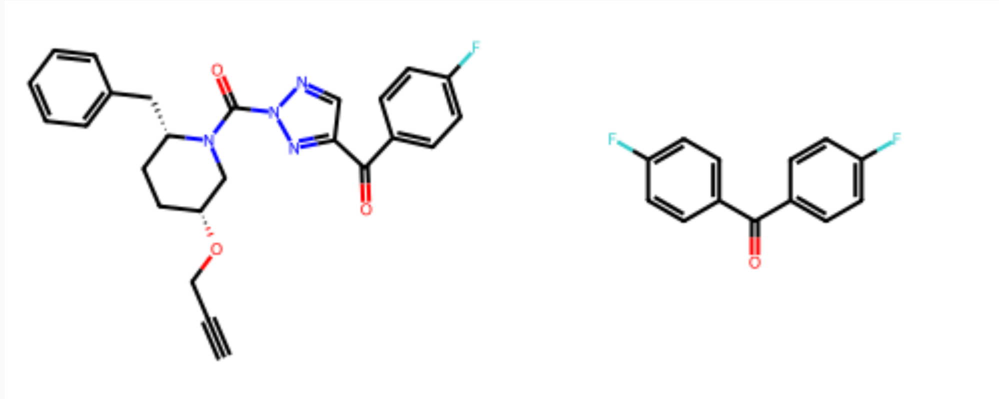
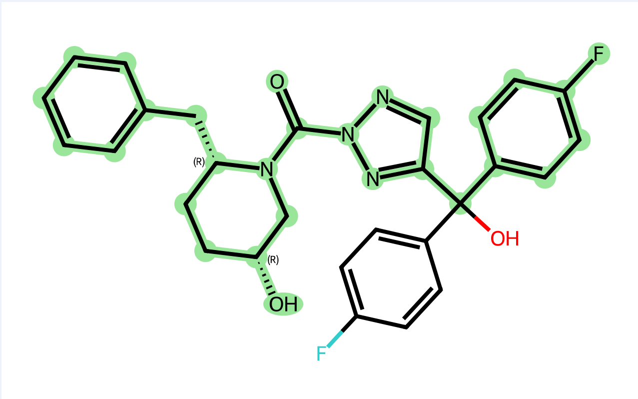
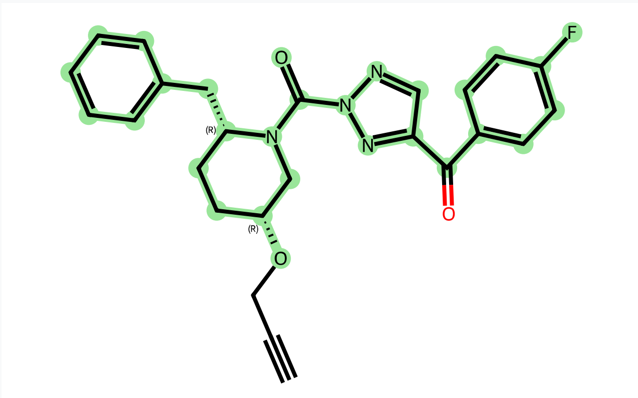
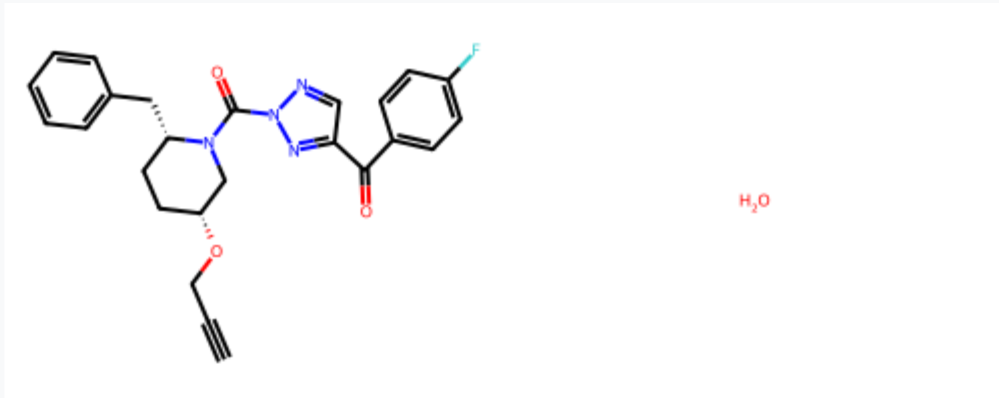
Top-10	-0.0857	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)Cl)C1.OC(c1ccc(F)cc1)(c1ccncc1)c1ccc(F)cc1		

MCS Match Group 2 (6 predictions) — best score -0.0438

MCS core (29 atoms):

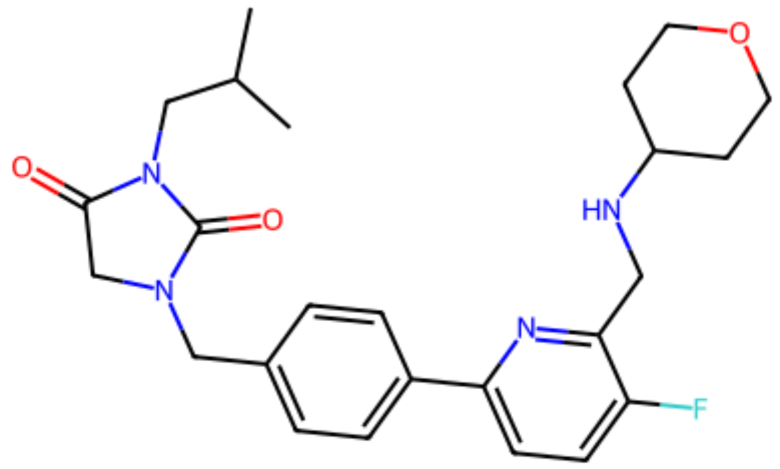


Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-2	-0.0438	C#CCBr.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cnn(C(=O)N2C[C@H](O)CC[C@@H]2Cc2ccccc2)n1		
Top-4	-0.0581	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(=O)c3ccc(F)cc3)n2)C1.[Mg+]c1ccc(F)cc1		
Top-5	-0.0600	C#CCBr.O=C(n1ncc(C(O)(c2ccc(F)cc2)c2ccc(F)cc2)n1)N1C[C@H](O)CC[C@@H]1Cc1ccccc1		

Top-6	-0.0679	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(=O)c3ccc(F)cc3)n2)C1.O=C(c1ccc(F)cc1)c1ccc(F)cc1		
Top-8	-0.0797	C#CCBr.O=C(Cl)Cl.OC(c1ccc(F)cc1)(c1ccc(F)cc1)c1cnn(C(=O)N2C[C@H](O)CC[C@@H]2Cc2ccccc2)n1		
Top-9	-0.0809	C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(=O)c3ccc(F)cc3)n2)C1.O		

Product 2

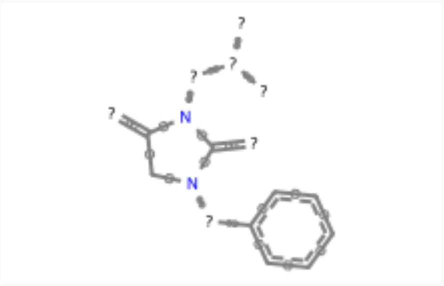
CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CNC4CCOCC4)n3)cc2)C1=O

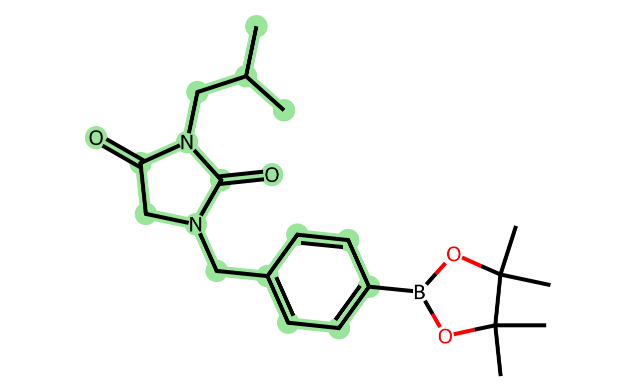
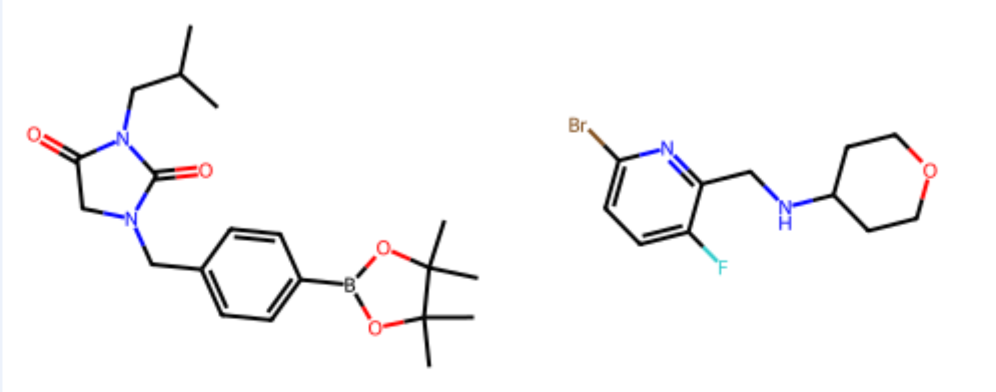
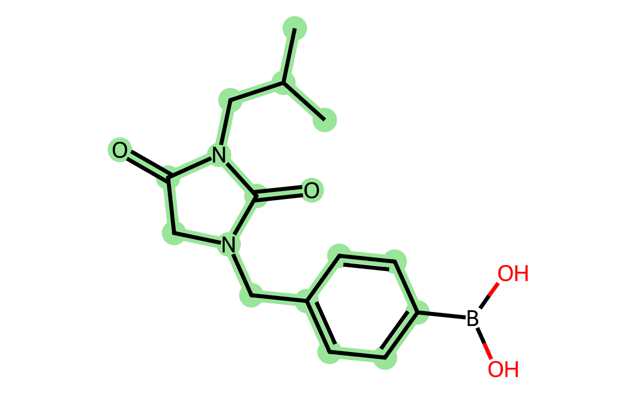
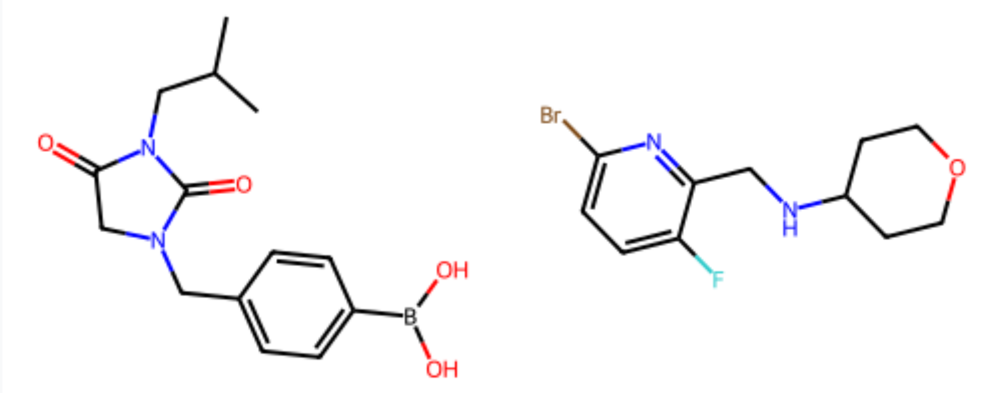


10 predictions → **2** MCS match groups

MCS Match Group 1 (2 predictions) — best score -0.0727

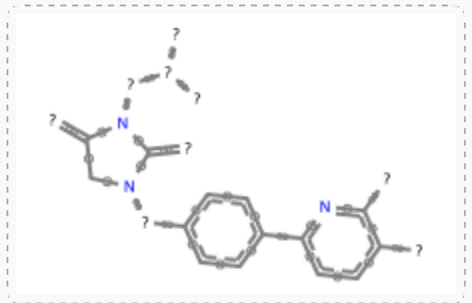
MCS core (18 atoms):

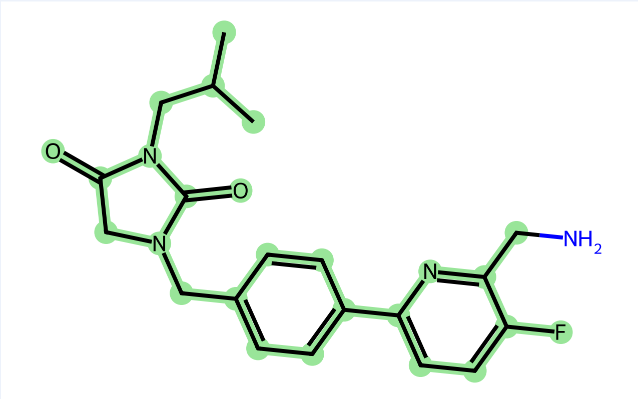
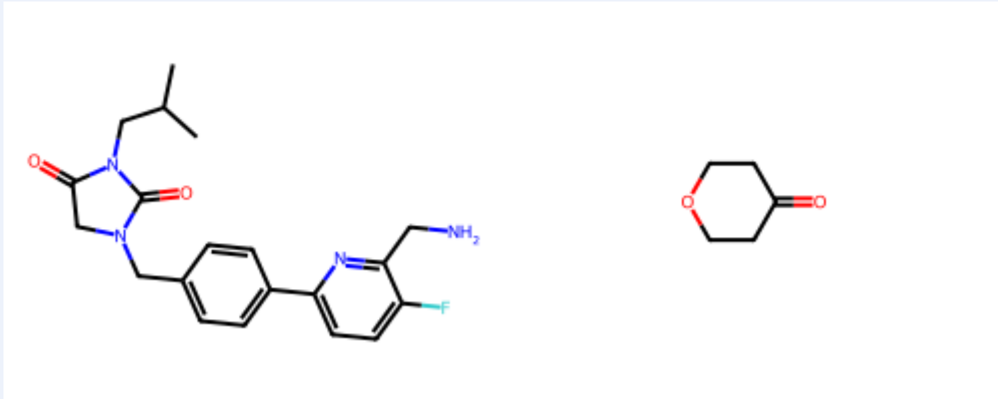
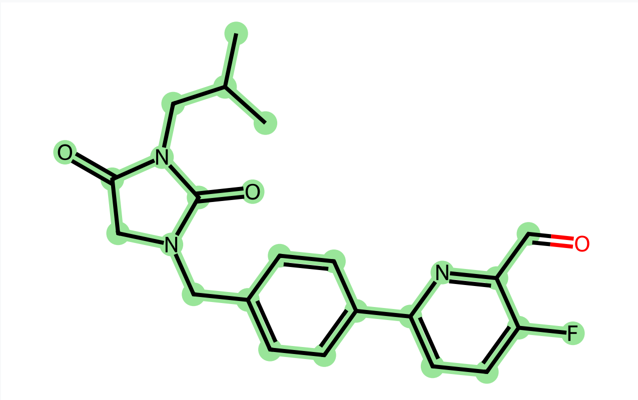
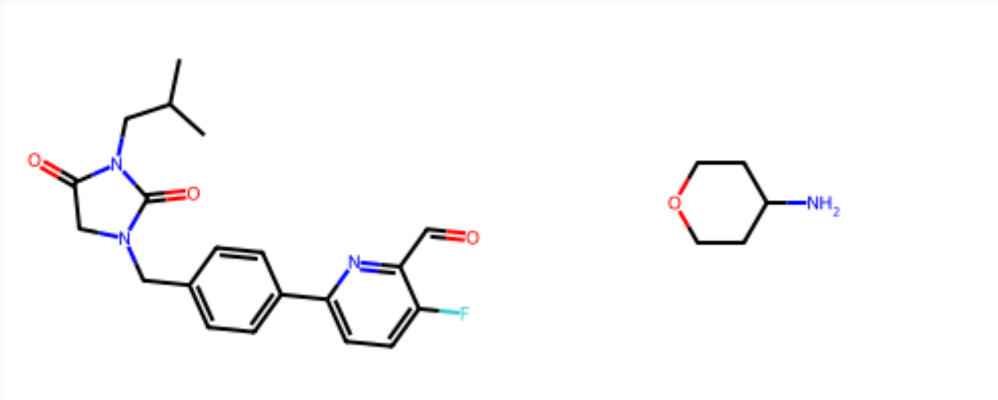
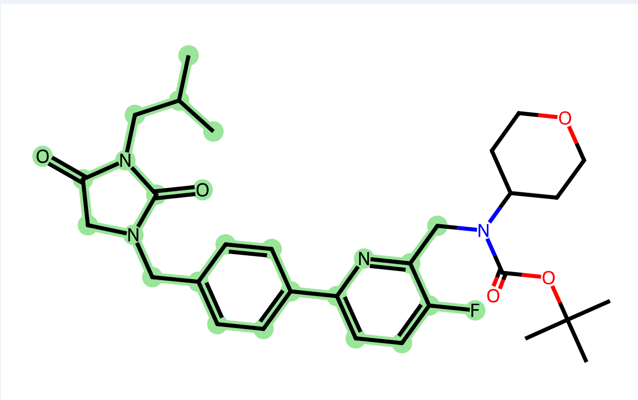
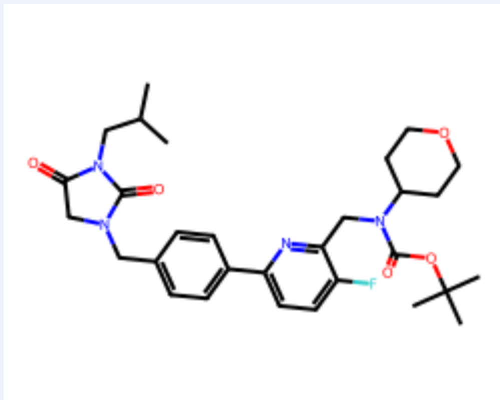


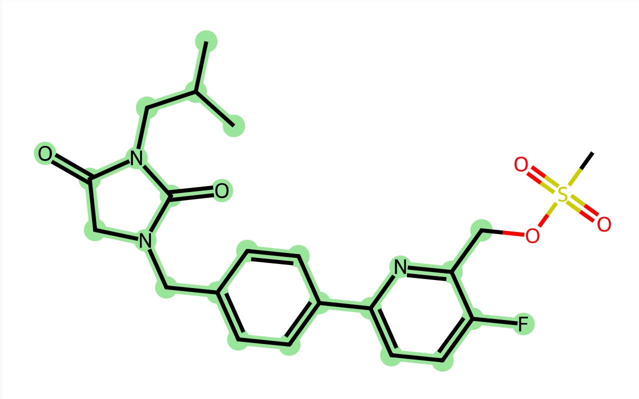
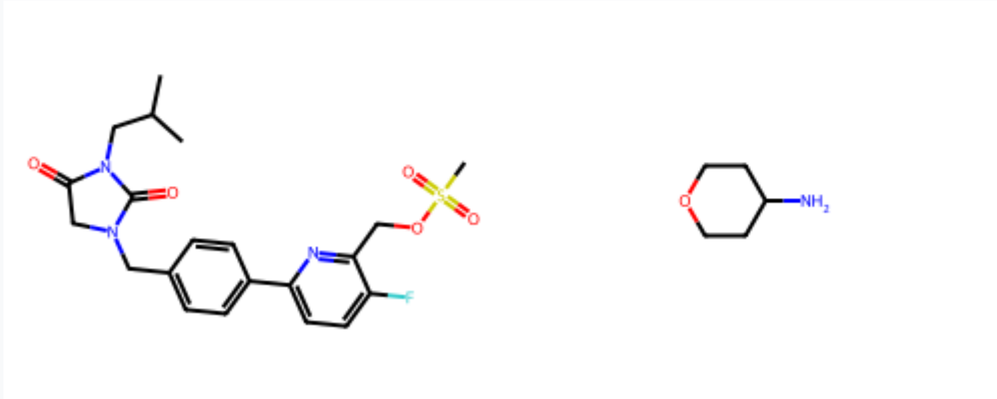
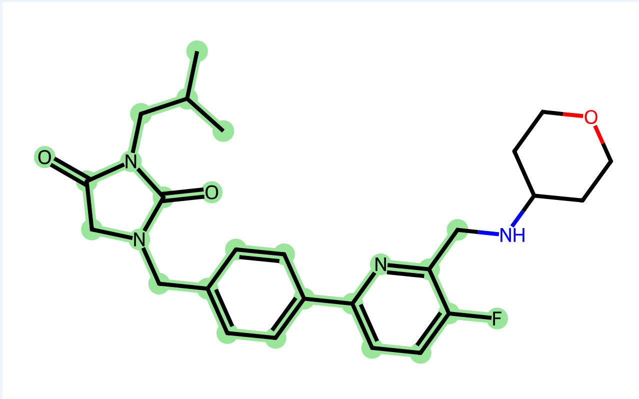
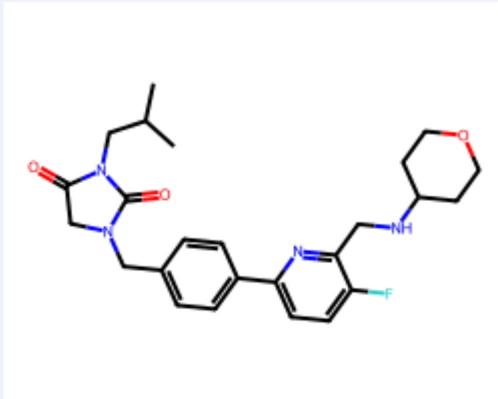
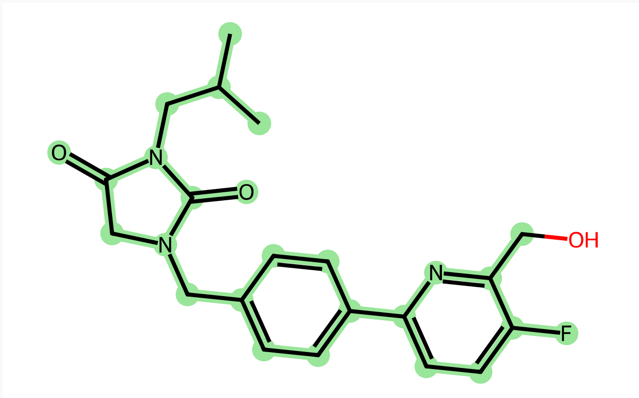
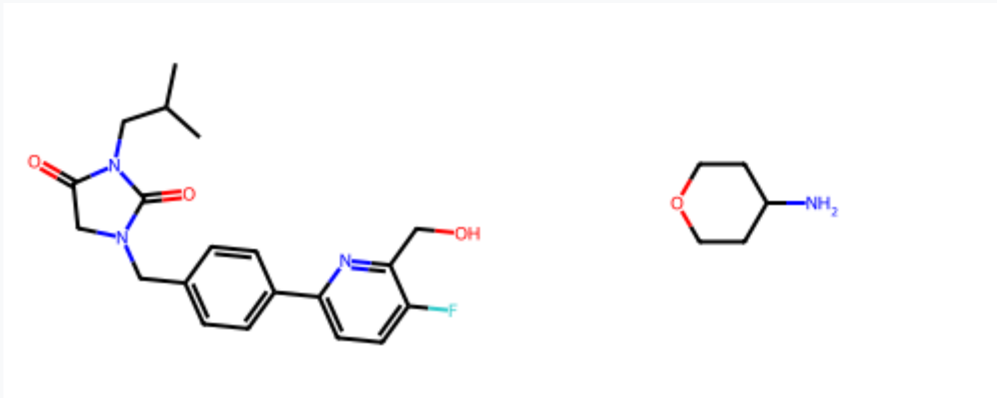
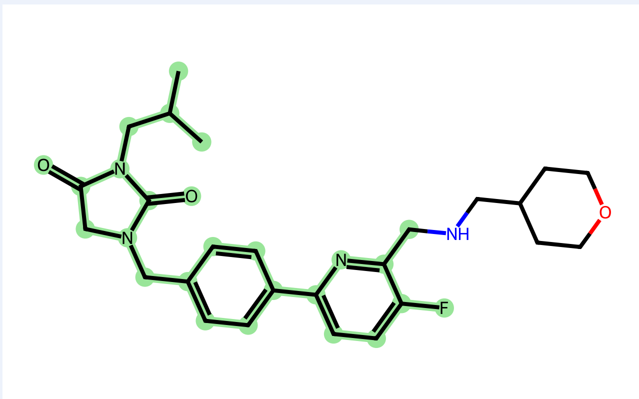
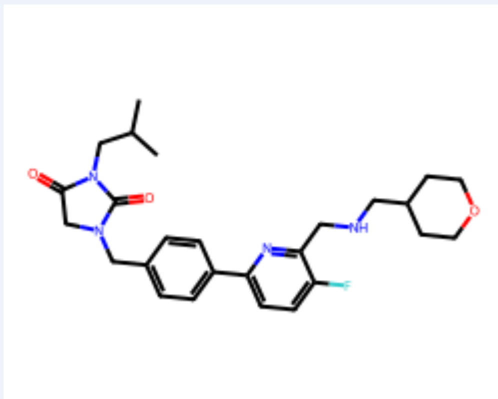
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-7	-0.0727	CC(C)CN1C(=O)CN(Cc2ccc(B3OC(C)(C)C(C)(C)O3)cc2)C1=O.Fc1ccc(Br)nc1CNC1CCOCC1		
Top-8	-0.0785	CC(C)CN1C(=O)CN(Cc2ccc(B(O)O)cc2)C1=O.Fc1ccc(Br)nc1CNC1CCOCC1		

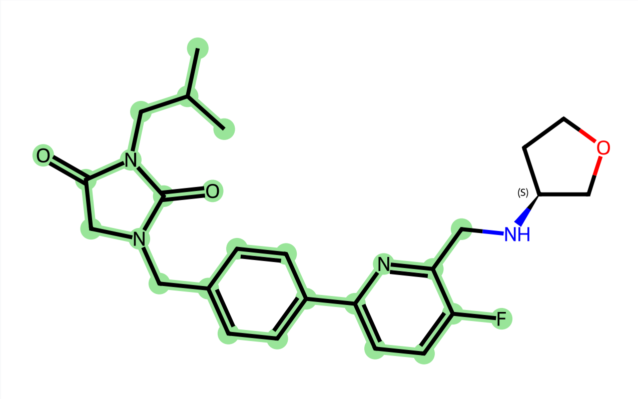
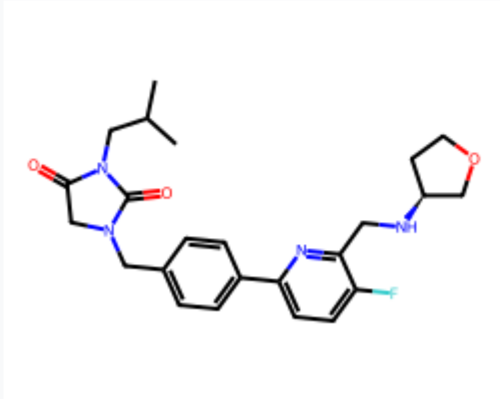
MCS Match Group 2 (8 predictions) — best score -0.0274

MCS core (26 atoms):



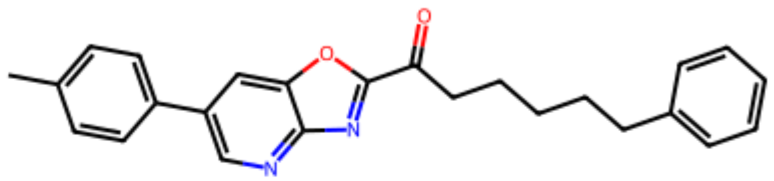
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0274	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CN)n3)cc2)C1=O.O=C1CCOCC1		
Top-2	-0.0329	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(C=O)n3)cc2)C1=O.NC1CCOCC1		
Top-3	-0.0344	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CN(C(=O)OC(C)(C)C)C4CCOCC4)n3)cc2)C1=O		

Top-4	-0.0388	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(COS(C)(=O)=O)n3)cc2)C1=O.NC1CCOCC1		
Top-5	-0.0393	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CNC4CCOCC4)n3)cc2)C1=O		
Top-6	-0.0514	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CO)n3)cc2)C1=O.NC1CCOCC1		
Top-9	-0.0992	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CNCC4CCOCC4)n3)cc2)C1=O		

Top-10	-0.1077	CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CN[C@H]4CCOC4)n3)cc2)C1=O		

Product 3

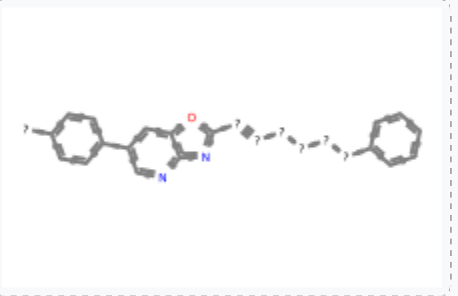
`Cc1ccc(-c2cnc3nc(C(=O)CCCCCc4ccccc4)oc3c2)cc1`

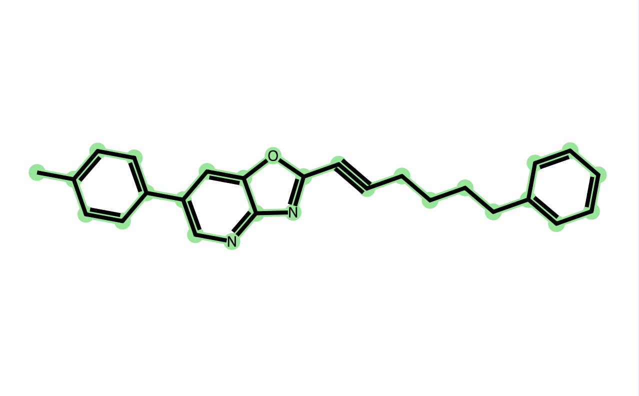
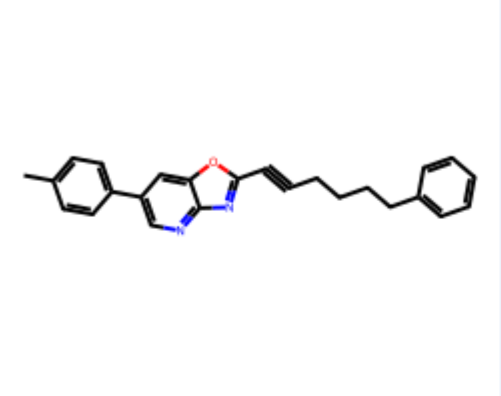
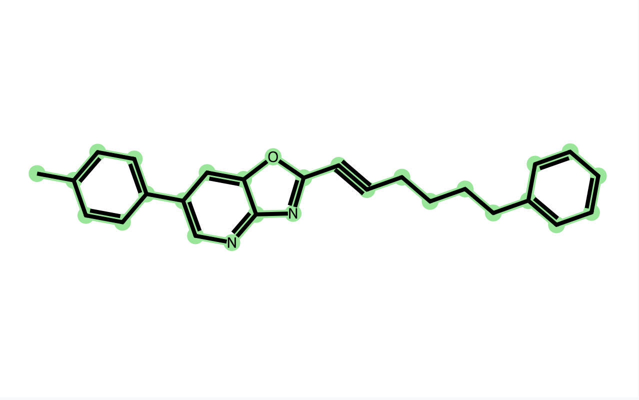
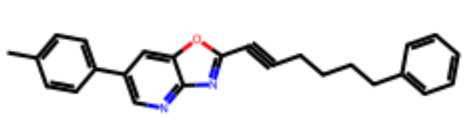
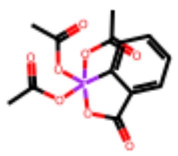
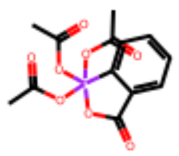


10 predictions → **3** MCS match groups

MCS Match Group 1 (2 predictions) — best score -0.0597

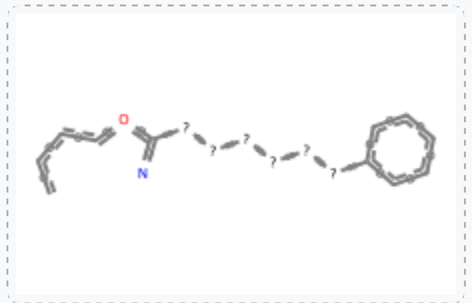
MCS core (28 atoms):

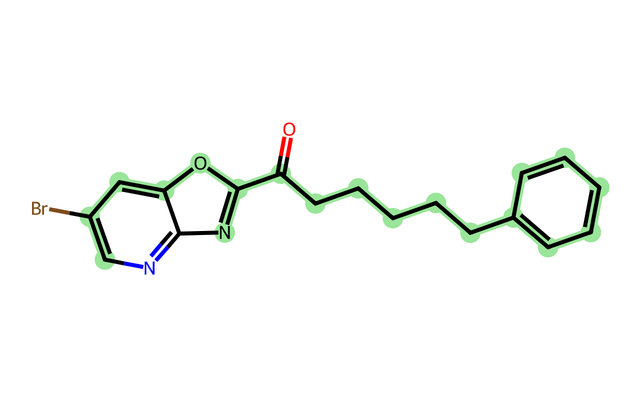
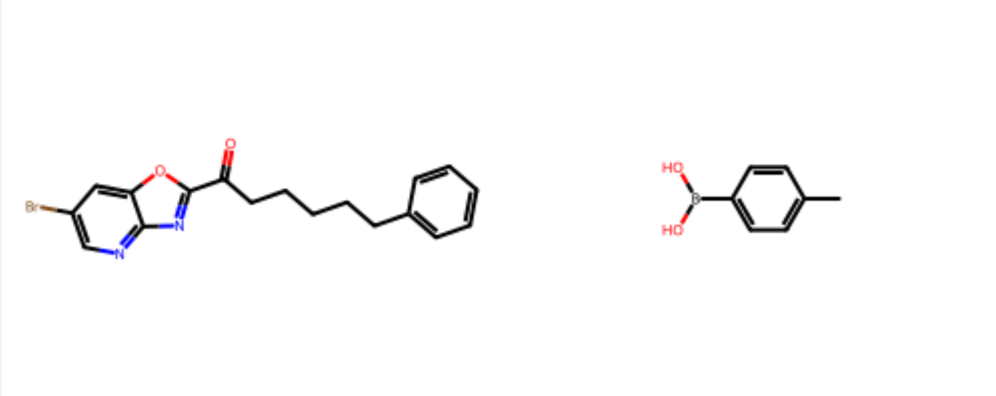
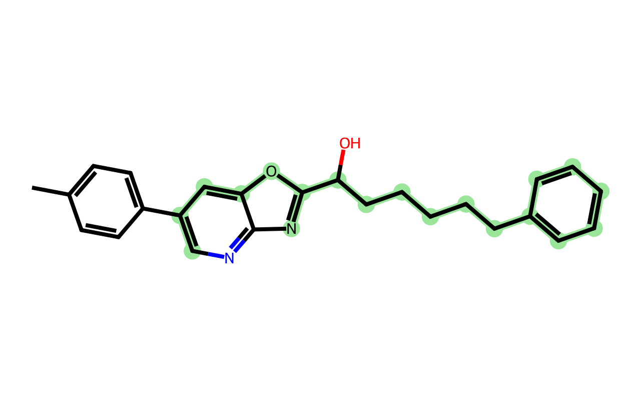
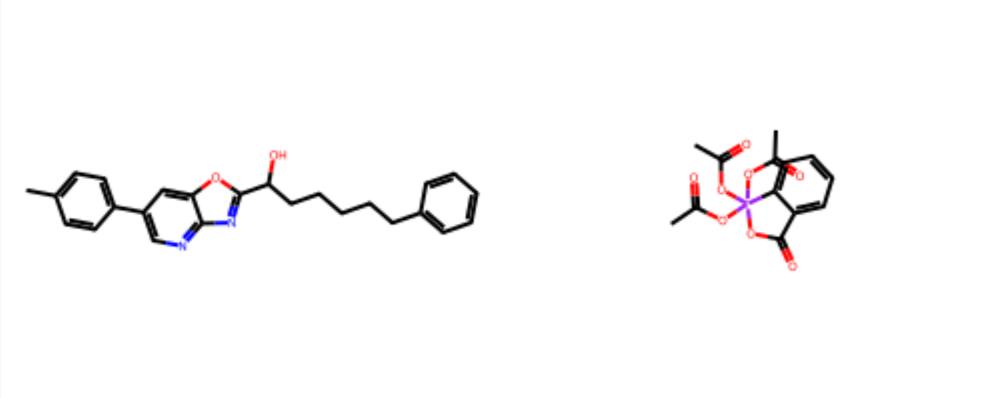
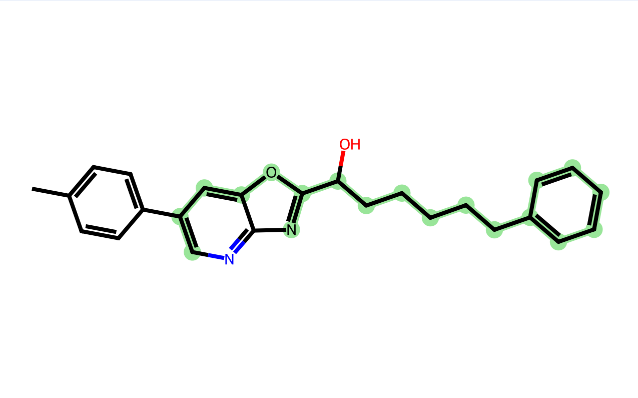
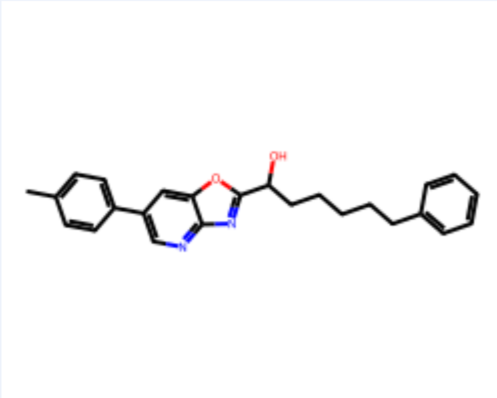


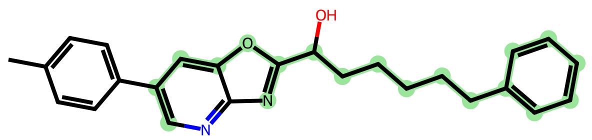
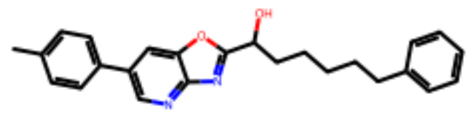
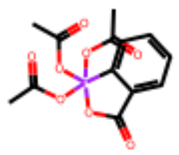
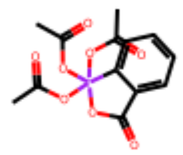
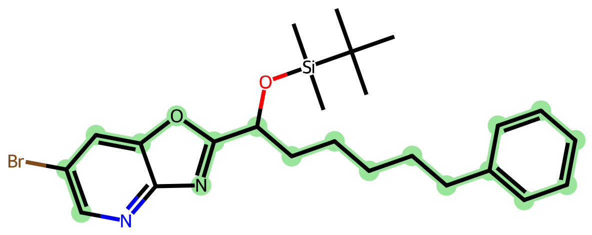
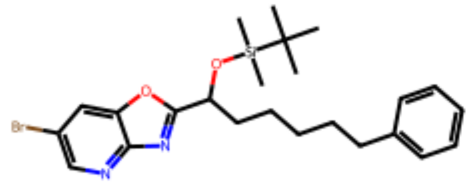
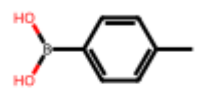
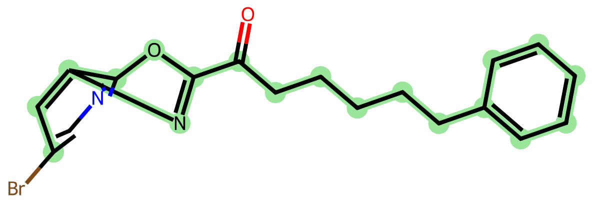
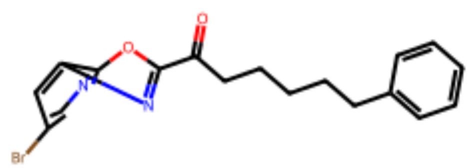
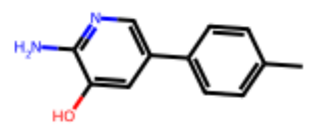
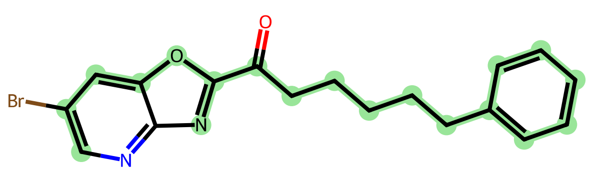
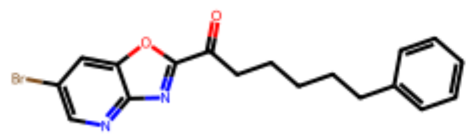
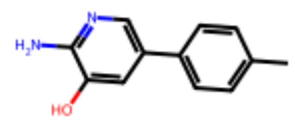
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants		
Top-5	-0.0597	Cc1ccc(-c2cnc3nc(C#CCCCCc4ccccc4)oc3c2)cc1				
Top-9	-0.1077	CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C#CCCCCc4ccccc4)oc3c2)cc1				

MCS Match Group 2 (7 predictions) — best score -0.0204

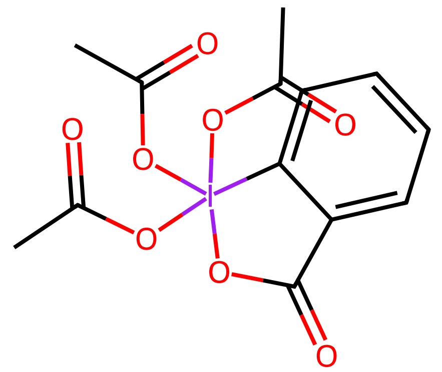
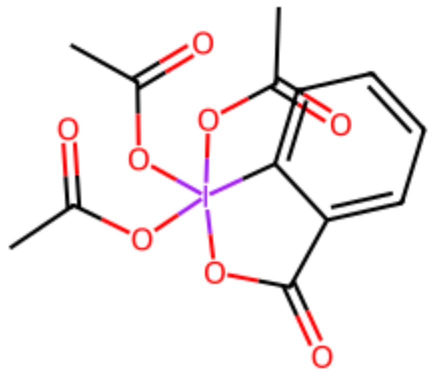
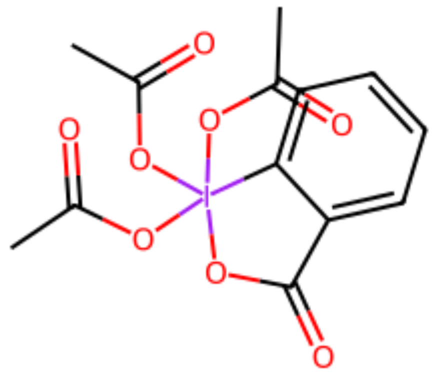
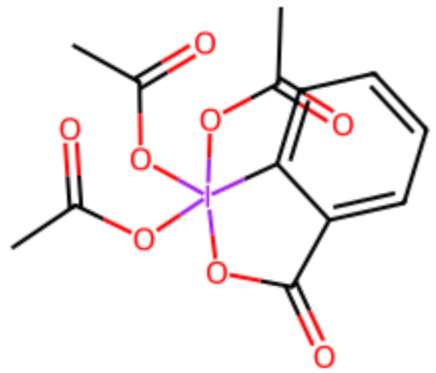
MCS core (19 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0204	Cc1ccc(B(O)O)cc1.O=C(CCCCCc1ccccc1)c1nc2ncc(Br)cc2o1		
Top-2	-0.0320	CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(O)CCCCc4ccccc4)oc3c2)cc1		
Top-3	-0.0423	Cc1ccc(-c2cnc3nc(C(O)CCCCc4ccccc4)oc3c2)cc1		

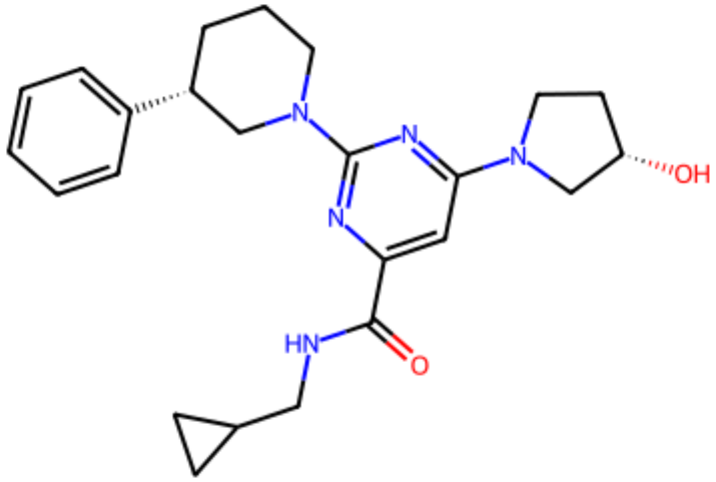
Top-4	-0.0595	CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(O)CCCCCc4ccccc4)oc3c2)cc1				
Top-6	-0.0625	CC(C)(C)[Si](C)(C)OC(CCCCCc1ccccc1)c1nc2ncc(Br)cc2o1.Cc1ccc(B(O)O)cc1				
Top-8	-0.1005	Cc1ccc(-c2cnc(N)c(O)c2)cc1.O=C(CCCCCc1ccccc1)c1nc2cc(Br)cnc2o1				
Top-10	-0.1086	Cc1ccc(-c2cnc(N)c(O)c2)cc1.O=C(CCCCCc1ccccc1)c1nc2ncc(Br)cc2o1				

MCS Match Group 3 (1 prediction) — best score -0.0804

Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-7	-0.0804	CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.CC(=O)OI1(OC(C)=O)(OC(C)=O)OC(=O)c2ccccc21.Cc1ccc(-c2cnc3nc(C(O)CCCCc4ccccc4)oc3c2)c		  

Product 4

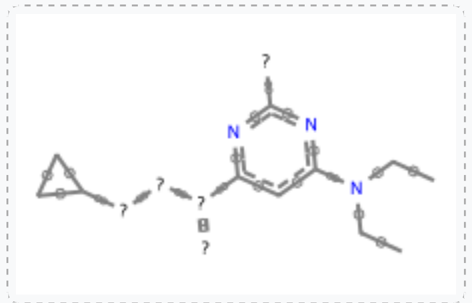
O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(N2CCC[C@@H](c3ccccc3)C2)n1

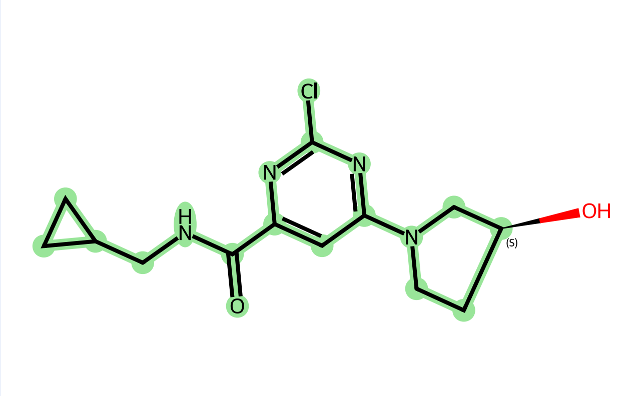
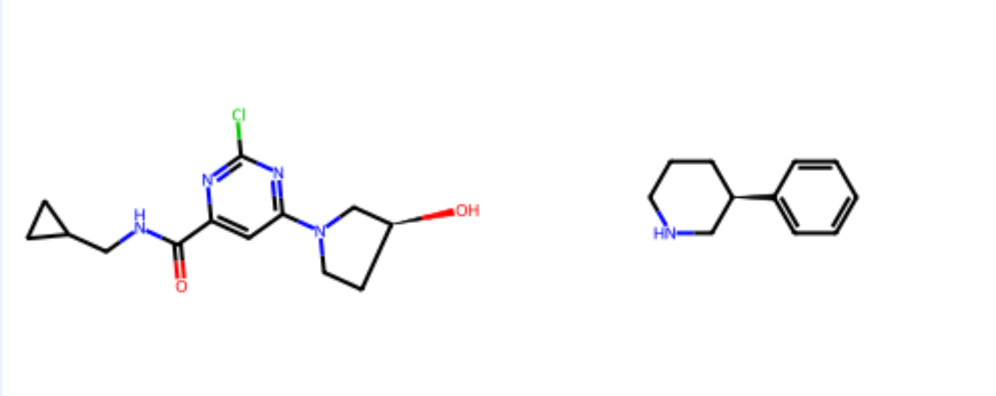
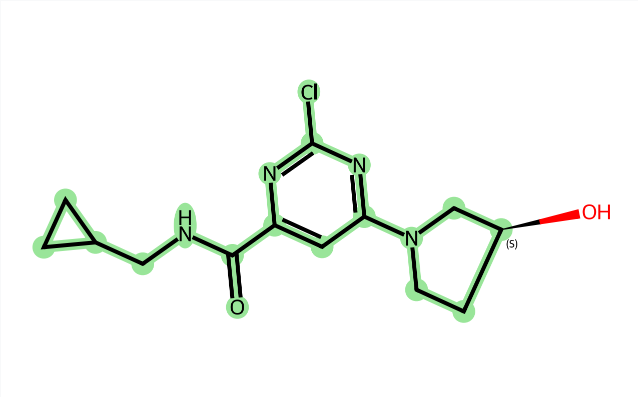
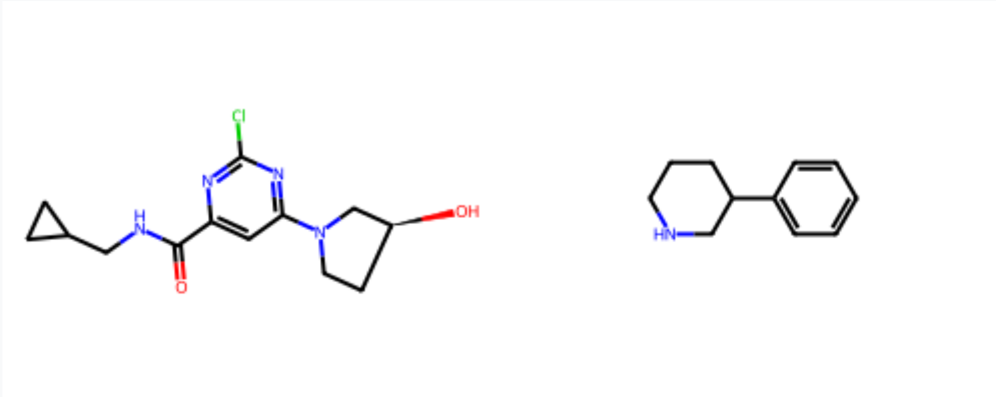
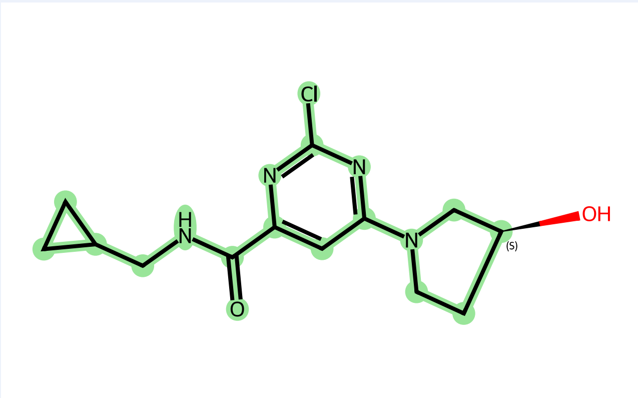
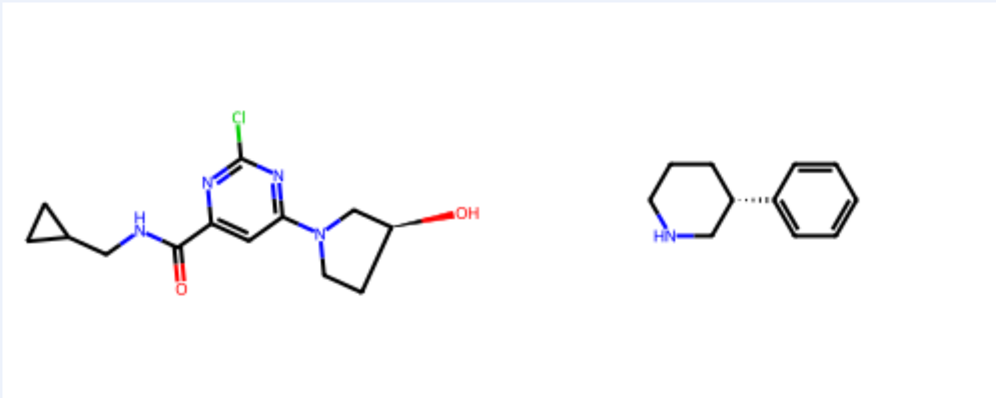


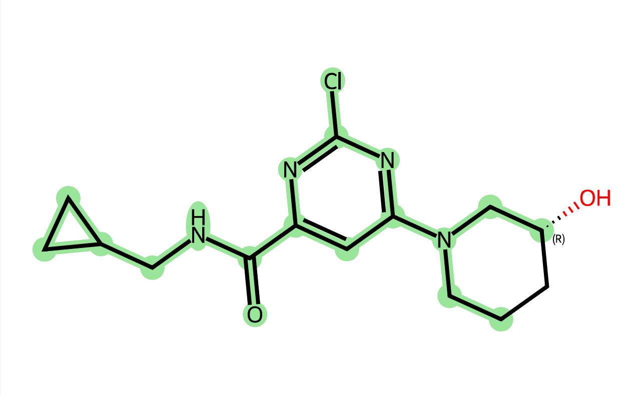
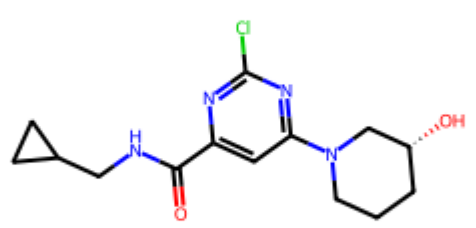
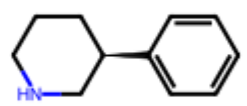
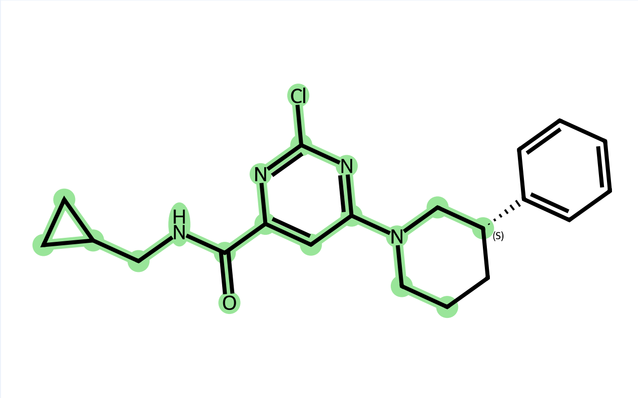
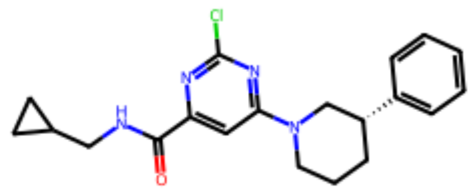
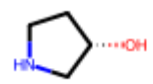
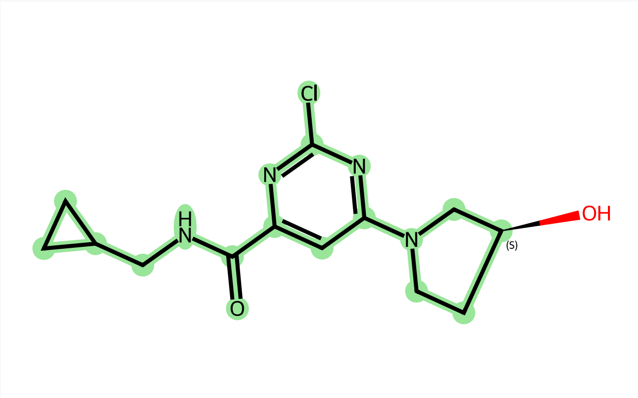
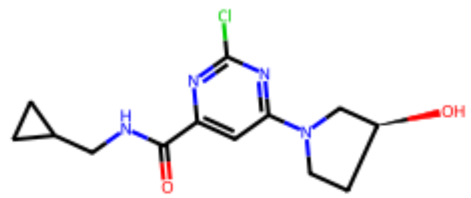
10 predictions → **2** MCS match groups

MCS Match Group 1 (6 predictions) — best score -0.0096

MCS core (19 atoms):

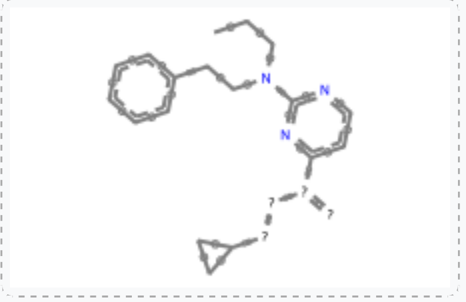


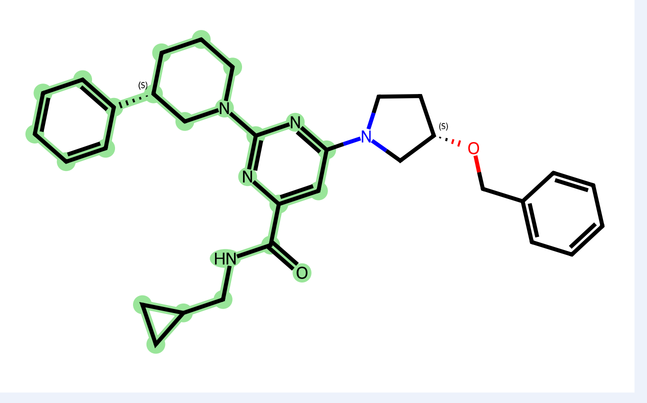
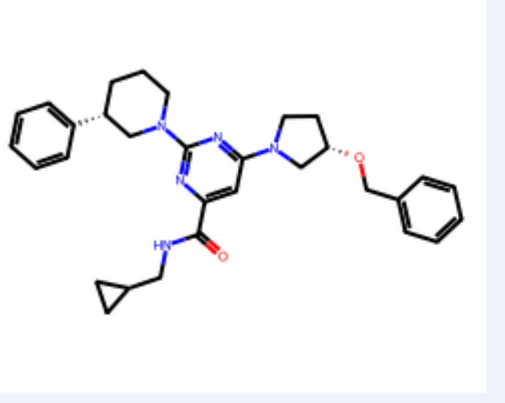
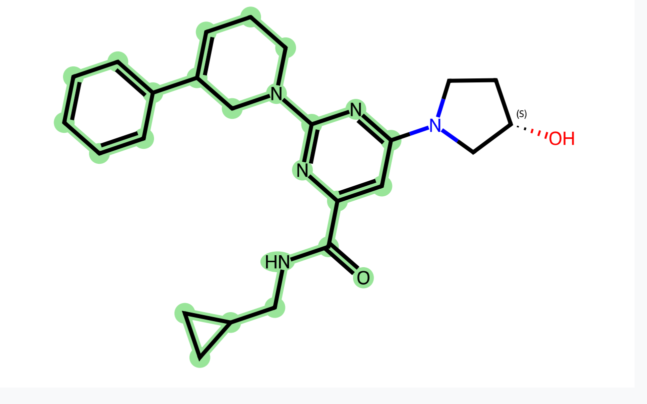
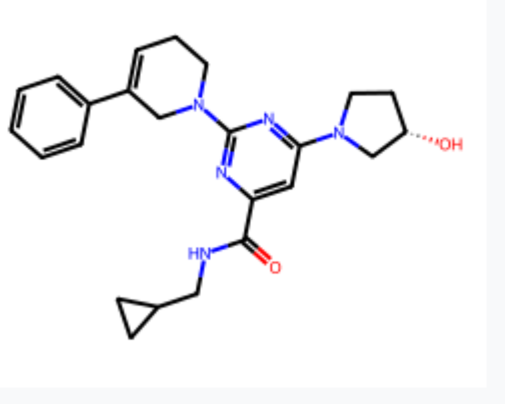
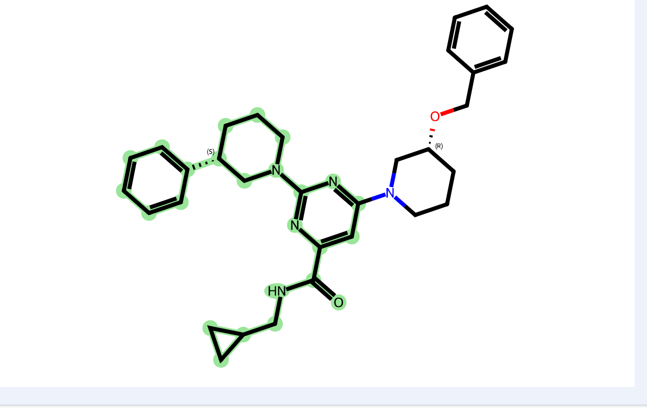
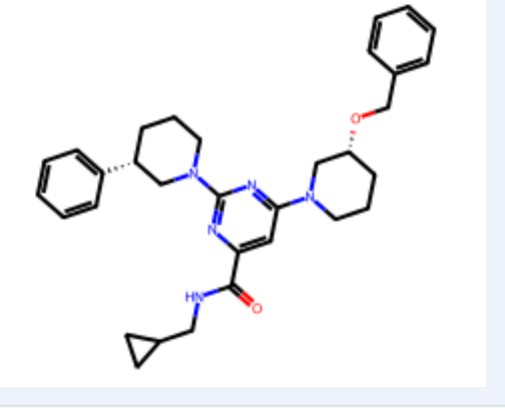
Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0096	O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.c1ccc([C@@H]2CCCNC2)cc1		
Top-3	-0.0417	O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.c1ccc(C2CCCNC2)cc1		
Top-4	-0.0489	O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1.c1ccc([C@H]2CCCNC2)cc1		

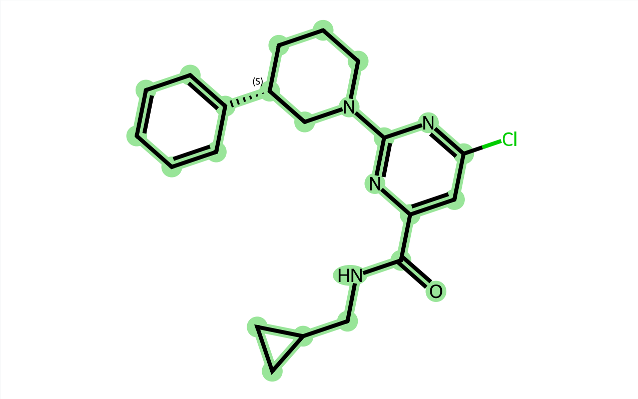
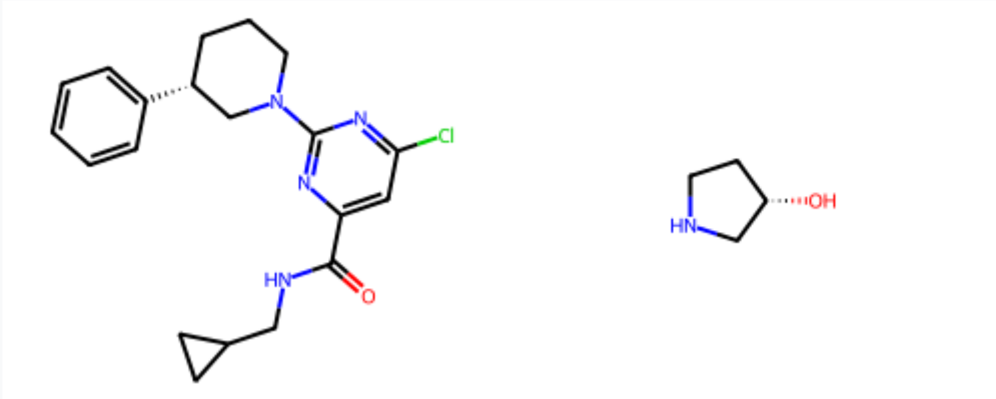
Top-6	-0.0651	O=C(NCC1CC1)c1cc(N2CCC[C@@H](O)C2)nc(Cl)n1.c1ccc([C@@H]2CCCNC2)cc1		 
Top-8	-0.0852	O=C(NCC1CC1)c1cc(N2CCC[C@@H](c3ccccc3)C2)nc(Cl)n1.O[C@H]1CCNC1		 
Top-10	-0.0962	O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(Cl)n1		

MCS Match Group 2 (4 predictions) — best score -0.0368

MCS core (25 atoms):

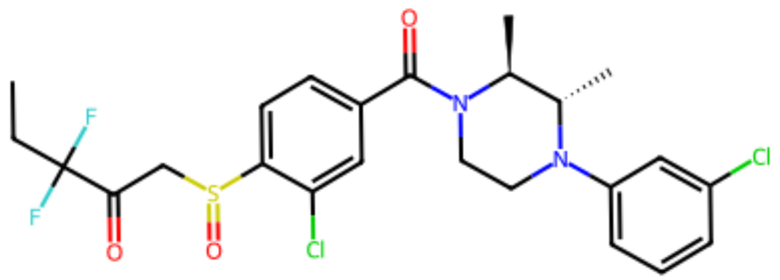


Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-2	-0.0368	O=C(NCC1CC1)c1cc(N2CC[C@H](OCc3ccccc3)C2)nc(N2CCC[C@@H](c3ccccc3)C2)n1		
Top-5	-0.0610	O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(N2CCC=C(c3ccccc3)C2)n1		
Top-7	-0.0656	O=C(NCC1CC1)c1cc(N2CCC[C@@H](OCc3ccccc3)C2)nc(N2CCC[C@@H](c3ccccc3)C2)n1		

Top-9	-0.0866	O=C(NCC1CC1)c1cc(Cl)nc(N2CCC[C@@H](c3ccccc3)C2)n1.O[C@H]1CCNC1		

Product 5

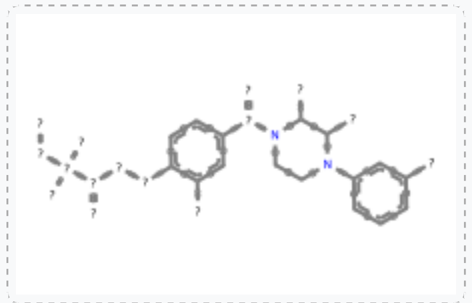
CCC(F)(F)C(=O)CS(=O)c1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl

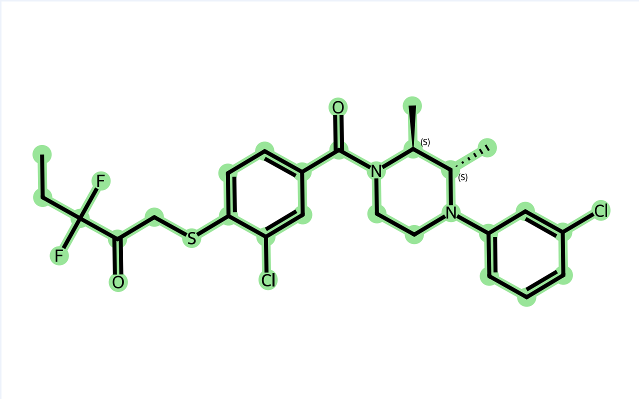
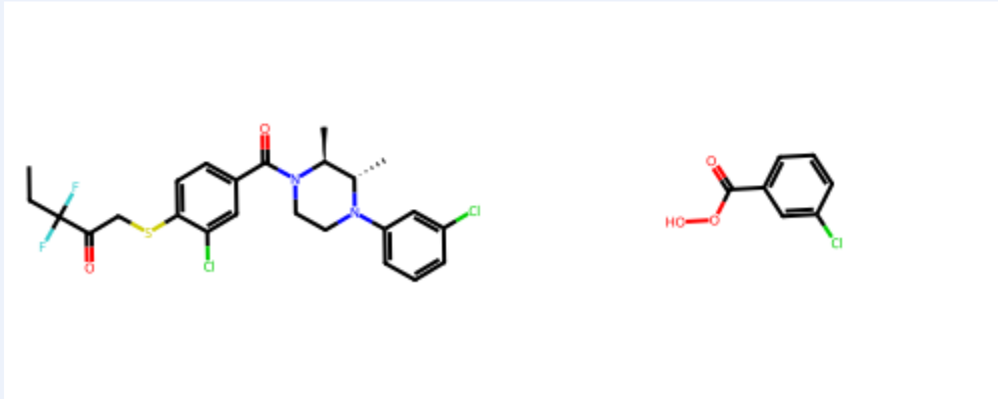
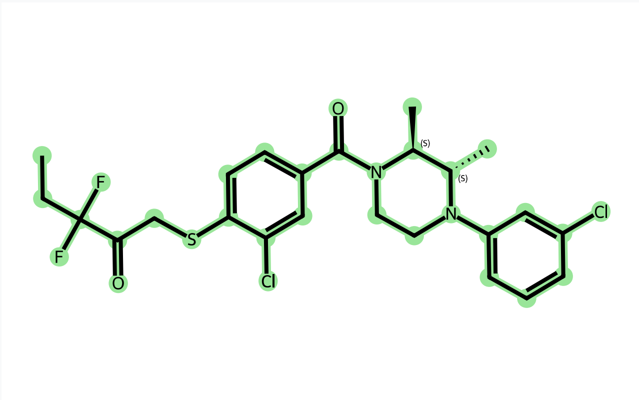
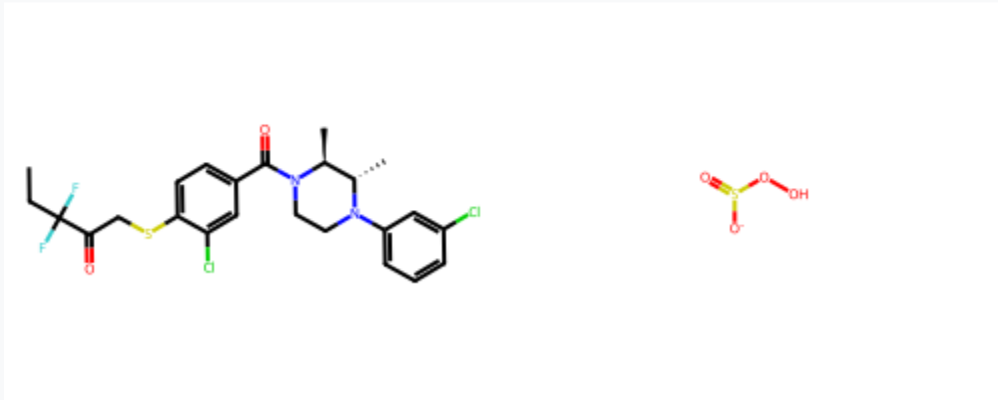
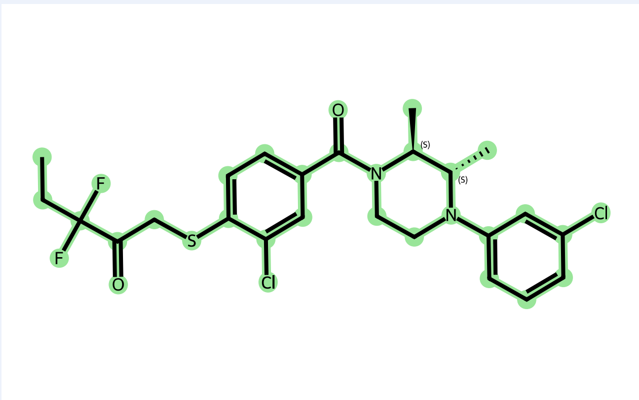
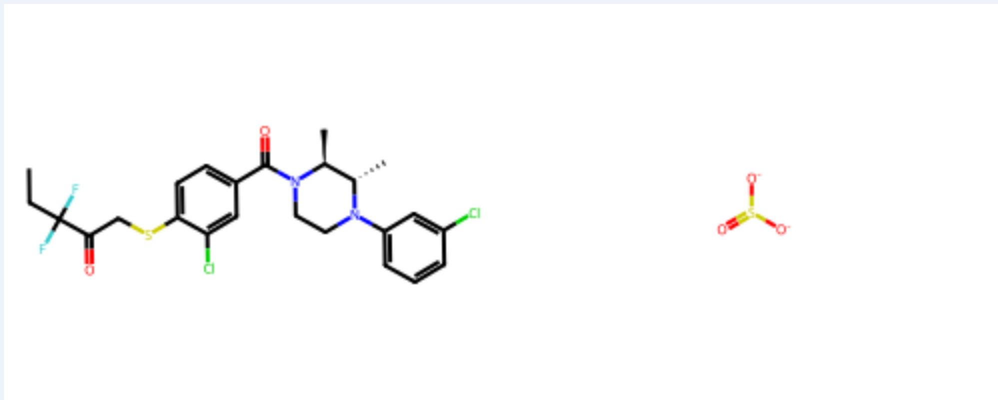


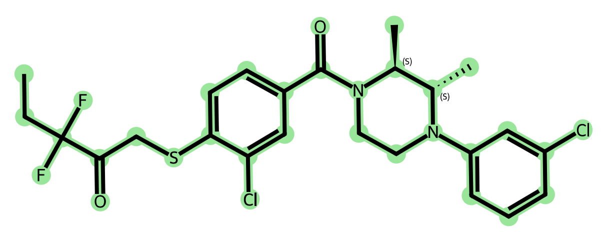
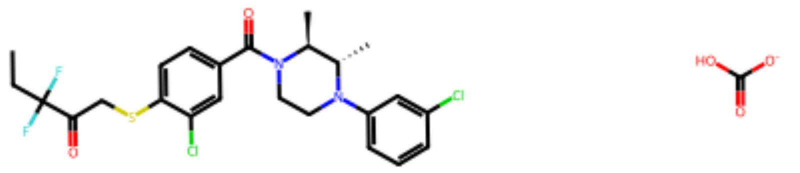
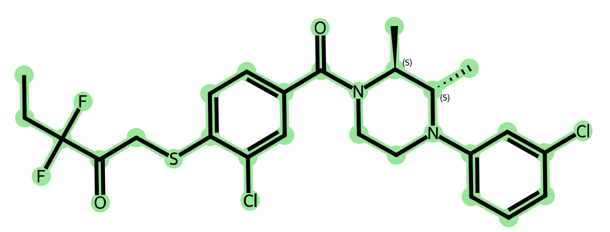
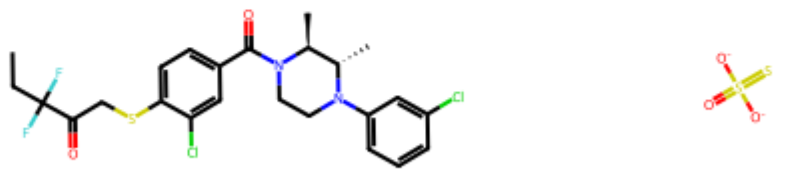
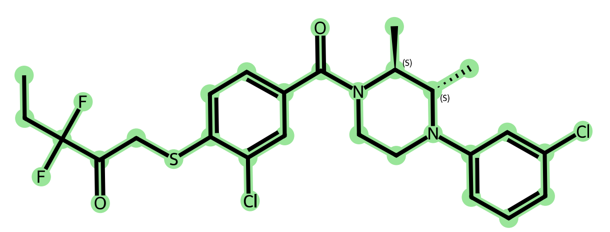
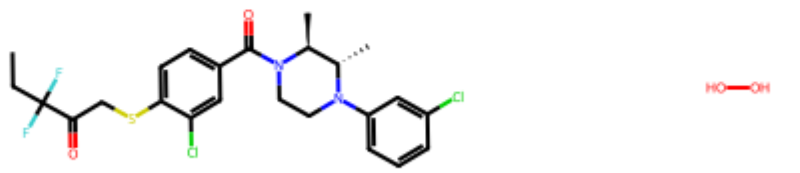
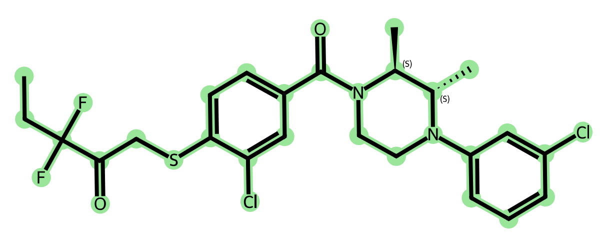
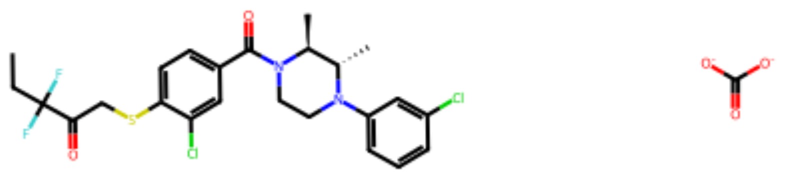
10 predictions → **1** MCS match group

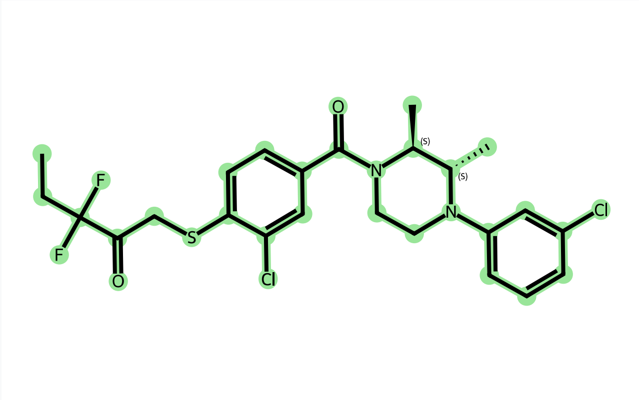
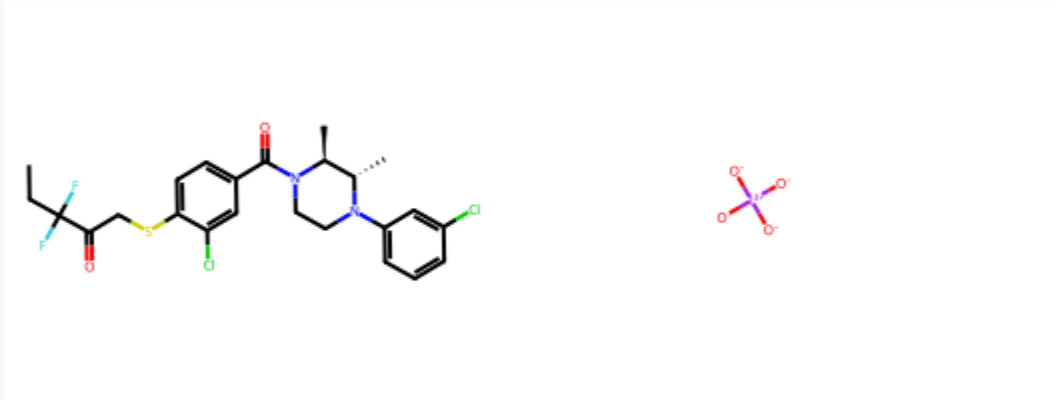
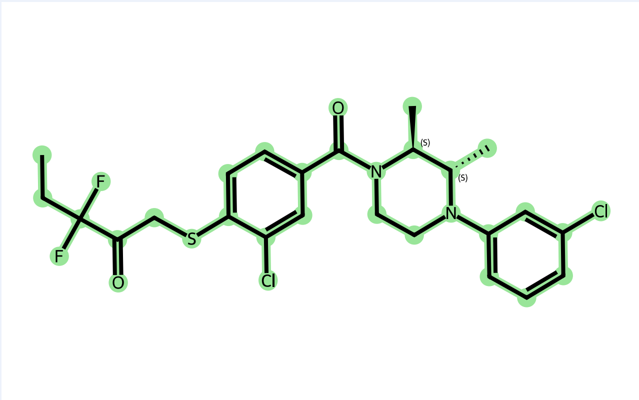
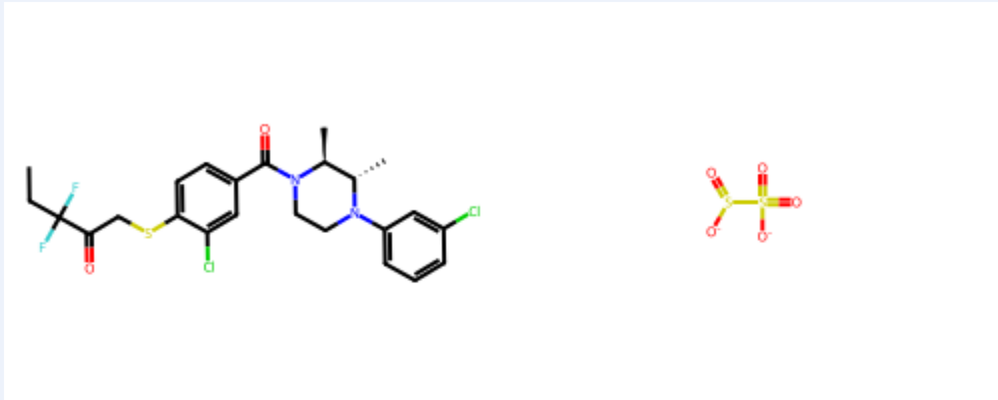
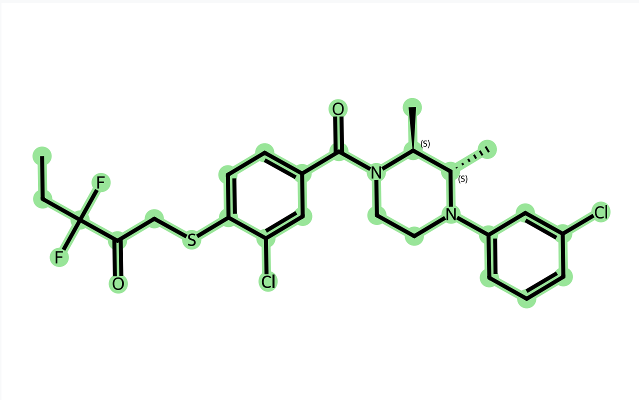
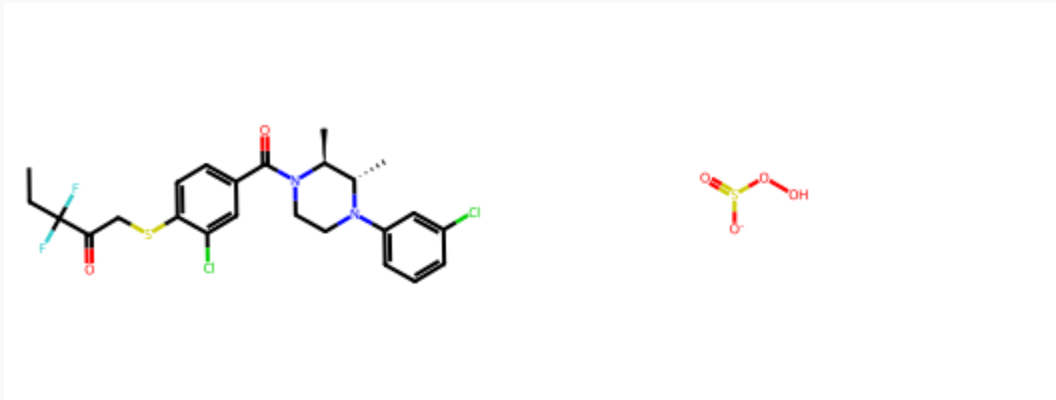
MCS Match Group 1 (10 predictions) — best score -0.0137

MCS core (33 atoms):



Rank	Score	Reactants SMILES	Largest Fragment (MCS highlighted)	All Reactants
Top-1	-0.0137	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=C(OO)c1cccc(Cl)c1		
Top-2	-0.0212	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=S([O-])OO		
Top-3	-0.0292	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=S([O-])[O-]		

Top-4	-0.0327	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=C([O-])O		
Top-5	-0.0346	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=S([O-])([O-])=S		
Top-6	-0.0413	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.OO		
Top-7	-0.0472	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=C([O-])[O-]		

Top-8	-0.0573	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.[Na+].[O-][I+3]([O-])([O-])[O-]			Na^+
Top-9	-0.0589	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=S([O-])S(=O)(=O)[O-]			
Top-10	-0.0606	CCC(F)(F)C(=O)CSc1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[C@@H]2C)cc1Cl.O=S([O-])OO.[K+]			K^+

📊 Summary

Product	# Predictions	# MCS Match Groups	Best Score
C#CCO[C@@H]1CC[C@H](Cc2ccccc2)N(C(=O)n2ncc(C(O)(c3ccc(F)cc3)…	10	2	0.0027
CC(C)CN1C(=O)CN(Cc2ccc(-c3ccc(F)c(CNC4CCOCC4)n3)cc2)C1=O	10	2	0.0274
Cc1ccc(-c2cnc3nc(C(=O)CCCCCc4ccccc4)oc3c2)cc1	10	3	0.0204
O=C(NCC1CC1)c1cc(N2CC[C@H](O)C2)nc(N2CCC[C@@H](c3ccccc3)C2)n…	10	2	0.0096
CCC(F)(F)C(=O)CS(=O)c1ccc(C(=O)N2CCN(c3cccc(Cl)c3)[C@@H](C)[…	10	1	0.0137